

AREZZO, Italy

WMO: 161720

Lat: 43.4597N

Lon: 11.8456E

Elev: 249

StdP: 98.37

Time Zone: 1.00 (EUC)

Period: 00-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	-3.1	-11.1	1.5	4.1	-9.0	1.8	2.9	9.3	8.6	8.5	8.3	0.1	100	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	35.0	20.4	33.2	20.0	31.9	19.6	22.7	30.0	21.7	29.1	20.9	28.3	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.7	25.3	19.4	14.6	24.4	18.5	13.8	23.6	68.0	29.6	64.3	29.2	61.3	28.2	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.8	5.6	-7.7	37.1	2.1	2.0	-9.2	38.5	-10.4	39.7	-11.6	40.8	-13.1	42.2		
(5)			-8.1	23.9	2.1	0.9	-9.6	24.5	-10.9	25.0	-12.1	25.6	-13.7	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	4.9	5.7	9.1	12.6	16.4	20.8	23.7	23.4	19.0	14.5	9.4	5.6		
	DBStd	7.32	3.22	3.25	3.24	2.82	2.70	3.04	2.45	2.48	3.03	3.02	3.60	3.56		
	HDD10.0	546	162	122	55	8	0	0	0	0	0	4	53	141		
	HDD18.3	2122	417	353	287	172	72	11	1	1	27	122	267	395		
	CDD10.0	1941	3	3	26	87	199	325	426	417	271	143	36	5		
	CDD18.3	476	0	0	0	1	13	85	168	159	48	3	0	0		
	CDH23.3	5736	0	0	0	32	246	1014	2052	1869	501	22	0	0		
	CDH26.7	2478	0	0	0	2	56	397	991	890	140	2	0	0		
(14) Wind	WSAvg	1.3	1.3	1.7	1.9	1.4	1.3	1.3	1.4	1.3	1.2	1.0	1.1	1.2		
(15) Precipitation	PrecAvg	770	65	61	65	60	62	49	40	49	63	78	86	79		
	PrecMax	1104	144	175	121	147	174	126	113	140	209	215	162	159		
	PrecMin	506	1	1	3	15	5	18	1	0	2	0	14	6		
	PrecStd	170	41	50	39	41	43	29	28	37	47	55	42	47		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	15.8	21.2	26.2	30.9	34.3	36.9	37.8	32.7	25.2	20.0	15.6		
		MCWB	11.2	10.6	11.5	14.6	18.0	20.4	20.5	20.8	18.7	16.3	14.5	12.6		
	2%	DB	13.0	13.8	19.2	24.0	28.1	32.8	35.1	35.2	30.2	23.4	18.0	14.0		
		MCWB	10.3	8.8	11.1	13.5	17.2	20.0	20.7	20.0	18.2	16.7	14.0	11.1		
	5%	DB	11.7	12.2	17.4	21.9	26.0	31.0	33.3	33.4	28.0	21.9	16.4	12.5		
		MCWB	9.2	8.6	10.7	13.1	16.3	19.8	20.2	20.0	17.9	16.1	13.2	10.0		
	10%	DB	10.1	11.1	15.7	19.9	24.1	29.0	31.9	31.5	26.1	20.2	15.1	11.0		
		MCWB	8.0	8.1	10.1	12.3	15.3	18.8	19.9	19.3	17.1	15.3	12.5	8.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.4	14.0	15.9	19.2	23.0	23.9	23.7	22.6	19.3	16.6	13.1		
		MCDB	13.7	13.8	18.9	22.3	27.1	29.7	31.2	32.0	26.2	22.3	18.4	15.0		
	2%	WB	10.9	10.3	12.6	14.7	18.1	21.5	22.7	22.5	20.6	18.2	14.7	11.8		
		MCDB	12.4	12.2	16.3	21.1	25.8	29.4	30.6	30.1	25.3	21.0	16.6	13.4		
	5%	WB	9.7	9.5	11.7	13.8	17.2	20.5	21.7	21.5	19.6	17.2	13.7	10.5		
		MCDB	11.2	11.3	15.3	20.0	24.4	28.5	29.4	28.7	25.0	20.1	15.8	11.9		
	10%	WB	8.5	8.6	10.8	12.9	16.2	19.5	20.9	20.7	18.5	16.3	12.7	9.4		
		MCDB	9.7	10.5	14.3	18.3	22.3	27.0	28.8	27.9	24.2	19.4	14.8	10.7		
(35) Mean Daily Temperature Range	MDBR		7.7	8.5	10.4	12.1	12.8	13.2	14.7	14.8	12.6	10.4	8.4	7.6		
	5% DB	MCDBR	8.3	10.0	14.7	16.8	16.8	16.3	17.5	18.0	15.4	12.2	9.4	8.5		
		MCWBR	5.9	6.3	7.6	7.8	7.2	6.0	5.7	5.8	5.9	6.4	6.1	6.0		
	5% WB	MCDBR	6.2	7.6	10.8	14.1	14.7	14.9	15.2	14.7	12.5	9.4	7.7	6.7		
		MCWBR	4.9	5.4	6.3	7.3	7.1	6.3	6.1	6.0	5.9	5.9	5.7	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.354	0.389	0.422	0.426	0.428	0.417	0.413	0.400	0.386	0.360	0.331		
	taud		2.439	2.354	2.269	2.193	2.221	2.260	2.293	2.322	2.338	2.363	2.376	2.455		
	Ebn at Noon		791	832	845	843	849	846	851	843	824	783	744	757		
	Edh at Noon		76	96	119	138	138	134	128	120	110	95	80	69		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.30	3.45	4.61	5.73	6.65	6.91	5.98	4.30	2.79	1.65	1.32		
	RadStd		0.18	0.32	0.51	0.46	0.56	0.37	0.39	0.41	0.26	0.28	0.18	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (18 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+74	N/A	(47)

Nomenclature: See separate page